

Python Programming Assignment

Section A – Basic Functions and Numbers

1. Even or Odd

```
def check_even_odd(n):  
    if n % 2 == 0:  
        print("Even")  
    else:  
        print("Odd")  
  
n = int(input("Enter a number: "))  
check_even_odd(n)
```

2. Positive, Negative or Zero

```
def check_number(n):  
    if n > 0:  
        print("Positive")  
    elif n < 0:  
        print("Negative")  
    else:  
        print("Zero")  
  
n = int(input("Enter a number: "))  
check_number(n)
```

3. Find Largest of Two Numbers

```
def find_largest(a, b):  
    if a > b:  
        return a  
    else:  
        return b  
  
a = int(input("Enter first number: "))  
b = int(input("Enter second number: "))
```

```
print("Largest =", find_largest(a, b))
```

4. Find Largest of Three Numbers

```
def find_largest(a, b, c):  
    if a >= b and a >= c:  
        return a  
    elif b >= a and b >= c:  
        return b  
    else:  
        return c  
  
a = int(input("Enter first number: "))  
b = int(input("Enter second number: "))  
c = int(input("Enter third number: "))  
  
print("Largest =", find_largest(a, b, c))
```

5. Sum of Natural Numbers

```
def sum_natural(n):  
    total = 0  
  
    for i in range(1, n + 1):  
        total += i  
  
    return total  
  
n = int(input("Enter n: "))  
print("Sum =", sum_natural(n))
```

6. Multiplication Table

```
def multiplication_table(n):  
    for i in range(1, 11):  
        print(n, "x", i, "=", n * i)  
  
n = int(input("Enter a number: "))  
multiplication_table(n)
```

7. Factorial Using Function

```
def factorial(n):  
    result = 1  
  
    for i in range(1, n + 1):  
        result *= i  
  
    return result  
  
n = int(input("Enter a number: "))  
print("Factorial =", factorial(n))
```

8. Count Digits

```
def count_digits(n):  
    if n == 0:  
        return 1  
  
    if n < 0:  
        n = -n  
  
    count = 0  
  
    while n > 0:  
        n //= 10  
        count += 1  
  
    return count  
  
n = int(input("Enter a number: "))  
print("Number of digits =", count_digits(n))
```

9. Reverse a Number

```
def reverse_number(n):  
    if n < 0:  
        n = -n  
        negative = True  
    else:  
        negative = False
```

```

reverse = 0

while n > 0:
    digit = n % 10
    reverse = reverse * 10 + digit
    n //= 10

if negative:
    reverse = -reverse

return reverse

n = int(input("Enter a number: "))
print("Reverse =", reverse_number(n))

```

10. Prime Number

```

def check_prime(n):
    if n < 2:
        return False

    for i in range(2, n):
        if n % i == 0:
            return False

    return True

n = int(input("Enter a number: "))

if check_prime(n):
    print("Prime")
else:
    print("Not Prime")

```

Section B – String-Based Assignments

11. Count Characters in a String

```

def count_characters(text):
    count = 0

    for char in text:

```

```
        count += 1

    return count

text = input("Enter a string: ")
print("Number of characters =", count_characters(text))
```

12. Count Vowels

```
def count_vowels(text):
    count = 0

    for char in text:
        if char.lower() in "aeiou":
            count += 1

    return count

text = input("Enter a string: ")
print("Number of vowels =", count_vowels(text))
```

13. Count Consonants

```
def count_consonants(text):
    count = 0

    for char in text:
        if char.isalpha() and char.lower() not in "aeiou":
            count += 1

    return count

text = input("Enter a string: ")
print("Number of consonants =", count_consonants(text))
```

14. Count Vowels and Consonants

```
def count_vowels_consonants(text):
    vowels = 0
    consonants = 0
```

```

for char in text:
    if char.isalpha():
        if char.lower() in "aeiou":
            vowels += 1
        else:
            consonants += 1

return vowels, consonants

text = input("Enter a string: ")

vowels, consonants = count_vowels_consonants(text)

print("Vowels      :", vowels)
print("Consonants :", consonants)

```

15. Reverse a String

```

def reverse_string(text):
    reversed_text = ""

    for char in text:
        reversed_text = char + reversed_text

    return reversed_text

text = input("Enter a string: ")

print("Using loop      :", reverse_string(text))
print("Using slicing :", text[::-1])

```

16. Check Palindrome String

```

def check_palindrome(text):
    reversed_text = ""

    for char in text:
        reversed_text = char + reversed_text

    if text == reversed_text:
        print("Palindrome")
    else:
        print("Not Palindrome")

```

```
text = input("Enter a string: ")
check_palindrome(text)
```

17. Count Words

```
def count_words(text):
    count = 0
    in_word = False

    for char in text:
        if char != " " and not in_word:
            count += 1
            in_word = True
        elif char == " ":
            in_word = False

    return count

text = input("Enter a sentence: ")
print("Number of words =", count_words(text))
```

18. Find Frequency of a Character

```
def character_frequency(text, ch):
    count = 0

    for char in text:
        if char == ch:
            count += 1

    return count

text = input("Enter text: ")
ch = input("Enter character: ")

print("Frequency =", character_frequency(text, ch))
```

19. Remove Spaces

```
def remove_spaces(text):  
    result = ""  
  
    for char in text:  
        if char != " "  
            result += char  
  
    return result  
  
text = input("Enter a string: ")  
print("Output:", remove_spaces(text))
```

20. Convert Lowercase to Uppercase

```
def convert_uppercase(text):  
    return text.upper()  
  
def convert_uppercase_loop(text):  
    result = ""  
  
    for char in text:  
        if 'a' <= char <= 'z':  
            result += chr(ord(char) - 32)  
        else:  
            result += char  
  
    return result  
  
text = input("Enter a string: ")  
  
print("Using upper() :", convert_uppercase(text))  
print("Using loop    :", convert_uppercase_loop(text))
```

Section C – String + Loop Problems

21. Count Uppercase, Lowercase, Digits and Spaces

```
def count_case(text):
    uppercase = 0
    lowercase = 0
    digits = 0
    spaces = 0

    for char in text:
        if char.isupper():
            uppercase += 1
        elif char.islower():
            lowercase += 1
        elif char.isdigit():
            digits += 1
        elif char == " ":
            spaces += 1

    print("Uppercase :", uppercase)
    print("Lowercase :", lowercase)
    print("Digits    :", digits)
    print("Spaces    :", spaces)

text = input("Enter a string: ")
count_case(text)
```

22. Find First Character

```
def first_character(text):
    for char in text:
        print("First character:", char)
        break

text = input("Enter a string: ")
first_character(text)
```

23. Find Last Character

```
def last_character(text):
    if text != "":
```

```
        print("Using indexing:", text[-1])

last = ""

for char in text:
    last = char

print("Using loop:", last)

text = input("Enter a string: ")
last_character(text)
```

24. Print Each Character

```
def display_characters(text):
    for char in text:
        print(char)

text = input("Enter a string: ")
display_characters(text)
```

25. Print Characters with Position

```
def display_position(text):
    for i in range(len(text)):
        print("Position", i, ":", text[i])

text = input("Enter a string: ")
display_position(text)
```

26. Remove Vowels

```
def remove_vowels(text):
    result = ""

    for char in text:
        if char.lower() not in "aeiou":
            result += char

    return result
```

```
text = input("Enter a string: ")
print("Output:", remove_vowels(text))
```

27. Find Longest Word

```
def find_longest_word(text):
    longest = ""
    current = ""

    for char in text:
        if char != " ":
            current += char
        else:
            if len(current) > len(longest):
                longest = current
            current = ""

    if len(current) > len(longest):
        longest = current

    return longest

text = input("Enter a sentence: ")
print("Longest word:", find_longest_word(text))
```

28. Count Occurrence of Each Vowel

```
def count_each_vowel(text):
    a = 0
    e = 0
    i = 0
    o = 0
    u = 0

    for char in text.lower():
        if char == "a":
            a += 1
        elif char == "e":
            e += 1
        elif char == "i":
            i += 1
        elif char == "o":
            o += 1
```

```

        elif char == "u":
            u += 1

    print("a =", a)
    print("e =", e)
    print("i =", i)
    print("o =", o)
    print("u =", u)

text = input("Enter a string: ")
count_each_vowel(text)

```

Section D – Loops, Patterns and Menus

29. Star Pattern

```

def star_pattern(n):
    for i in range(1, n + 1):
        for j in range(i):
            print("*", end=" ")
        print()

n = int(input("Enter number of rows: "))
star_pattern(n)

```

30. Number Pattern

```

def number_pattern(n):
    for i in range(1, n + 1):
        for j in range(1, i + 1):
            print(j, end=" ")
        print()

n = int(input("Enter number of rows: "))
number_pattern(n)

```